Metric Space

Labix

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1 Metric Spaces

Metric spaces lies in between analysis and topology. On one hand, they are the general setting to discuss the notion of length, and hence sequences and convergence. On the other hand, almost all of the notions of metric spaces are subsumed by topological spaces with a relative notion of closeness. Hence they form a large class of examples of topological spaces and its properties, while being easy to visualize.

1.1 Basic Definitions

Definition 1.1.1 (Metric)
Let X be a set. A metric on X is a function $d : X \times X \to \mathbb{R}$ such that the following are true.
- For all $x, y \in X$, we have $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
- $d(x, y) = d(y, x)$
- $d(x, y) \leq d(x, z) + d(z, y)$

Definition 1.1.2 (Metric Space)
A metric space is an oredered pair (X, d) where X is a set and d is a metric on X.

Example 1.1.3 (The Standard Euclidean Metric on $\mathbb{R}$)
The function $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by

$$d(x, y) = \sqrt{x^2 + y^2}$$

defines a metric on $\mathbb{R}$, called the standard Euclidean metric.

Example 1.1.4 (Exotic Metrics in $\mathbb{R}^2$)
The following functions define metrics on $\mathbb{R}^2$.
- Define the standard Euclidean metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- Define the taxicab metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|$$

- Define the French railway metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \begin{cases} \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} & \text{if } x_2 y_1 = y_2 x_1 \\ \sqrt{x_1^2 + x_2^2} + \sqrt{y_1^2 + y_2^2} & \text{otherwise} \end{cases}$$

- Define the jungle river metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \begin{cases} |y_1 - y_2| & \text{if } x_1 = x_2 \\ |y_1| + |x_1 - x_2| + |y_2| & \text{otherwise} \end{cases}$$

- Define the box metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \max\{|x_1 - x_2|, |y_1, y_2|\}$$

Example 1.1.5 (The Discrete Metric)
Let X be a set. Define the discrete metric on X to be the function $d : X \times X \to \mathbb{R}$ given by

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{otherwise} \end{cases}$$

1.2 New Metrics from Old

Lemma 1.2.1 (Metric Subspace)
Let (X,d) be a metric space. Let $U \subseteq X$ be a subset. Then $(U, d|_{U\times U})$ is also a metric space

Proof
$d|_{U\times U}$ has the same properties as d. ■

Definition 1.2.2 (The p Metric) Let (X,d_X) and (Y,d_Y) be metric spaces. Let $p \in [1,\infty)$. Define the p metric on $X \times Y$ to be the map $d_p : (X\times Y)\times(X\times Y) \to \mathbb{R}$ given by

$$d_p((x_1,y_1),(x_2,y_2)) = (d_X(x_1,x_2)^p + d_Y(y_1,y_2)^p)^{1/p}$$

Proposition 1.2.3 Let (X,d_X) and (Y,d_Y) be metric spaces. Let $p \in [1,\infty)$. Then d_p is a metric on $X \times Y$.

Example 1.2.4
Let $n \in \mathbb{N}$. Then the function $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined by

$$d((x_1,\dots,x_n),(y_1,\dots,y_n)) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

is a metric in $\mathbb{R}^n$, called the standard Euclidean metric of $\mathbb{R}^n$.

1.3 Size and Distance of Subsets

Definition 1.3.1 (Open Balls)
Let (X,d) be a metric space. Let $x \in X$. Let $r > 0$. Define the open ball of radius r at x to be

$$B_r(x) = \{y \in X \mid d(x,y) < r\}$$

Definition 1.3.2 (Bounded Subsets)
Let X be a metric space. Let $U \subseteq X$ be a subset. We say that U is bounded if there exists $x \in X$ and $r > 0$ such that $U \subseteq B_r(x)$.

Definition 1.3.3 (Distance between two Subsets)
Let X be a metric space. Let $U, V \subseteq X$ be subsets. Define the distance between U and V to be

$$\text{dist}(U,V) = \inf\{d(u,v) \mid u \in U, v \in V)\}$$

1.4 Distance Preserving Maps

Definition 1.4.1 (Distance Preserving Maps)
Let (X,d_X) and (Y,d_Y) be metric spaces. Let $f : X \to Y$ be a function. We say that f is distance preserving if for any $p, q \in X$, we have

$$d_Y(f(p), f(q)) = d_X(p,q)$$

Definition 1.4.2 (Isometries) Let (X,d_1) and (X,d_2) be metric spaces. Let $f : X \to Y$ be a distance preserving map. We say that f is an isometry if there exists a distance preserving map $g : Y \to X$ such that

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g = \text{id}_Y$$

It is also common to use the term isometry to mean a distance preserving function and refer to our version of isometry as bijective isometries.

Lemma 1.4.3 Let (X, d_1) and (X, d_2) be metric spaces. Let $f : X \to Y$ be a distance preserving map. Then f is an isometry if and only if f is bijective.

Proof
Clearly, if f is an isometry then f has an inverse function. Conversely, suppose that f is bijective. Then we have

$$d_X(f^{-1}(p), f^{-1}(q)) = d_Y(f(f^{-1}(p)), f(f^{-1}(q))) = d_Y(p, q)$$

for any $p, q \in Y$ since f is distance preserving. Hence f^{-1} is distance preserving. ∎

Definition 1.4.4 (The Set of Isometries of a Metric Space)
Let (X, d) be a metric space. Define the set of isometries of X to be

$$\text{Isom}(X, d) = \{f : X \to X \mid f \text{ is an isometry }\}$$

2 Topological Properties

2.1 Open and Closed Subsets

Definition 2.1.1 (Open and Closed Sets)
Let X be a metric space. Let $U \subseteq X$.

- We say that U is an open subset of X if for all $x \in U$, there exists $r > 0$ such that $B_r(x) \subseteq U$.
- We say that U is closed if $X \setminus U$ is open.

Example 2.1.2
Consider the following subsets of $\mathbb{R}$.

- $(0, 1)$ is open in $\mathbb{R}$.
- $[0, 1]$ is closed in $\mathbb{R}$.
- $(0, 1]$ and $[0, 1)$ are neither open nor closed in $\mathbb{R}$.

Example 2.1.3
Let $[0, 1)$ inherit a metric subspace structure from $\mathbb{R}$.

- $[1/2, 1)$ is closed in $[0, 1)$.
- $[0, 1)$ is both open and closed in $[0, 1)$.

Lemma 2.1.4
Let X be a metric space. Let $x \in X$. Let $r > 0$. Then $B_r(x)$ is an open subset of X.

Proof Let $B_r(a)$ be our open ball. Let $x \in B_r(a)$. Then

$$B_{(r-d(x,a))/2}(x) \subseteq B_r(a)$$

thus we are done. ■

Proposition 2.1.5
Let X be a metric space. Then the following are true.

- Let $\{U_i \mid i \in I\}$ be a collection of open subsets of X. Then $\bigcup_{i \in I} U_i$ is an open subset of X.
- Let $U_1, \dots, U_n$ be open subsets of X. Then $\bigcap_{i=1}^n U_i$ is an open subset of X.
- Let $\{U_i \mid i \in I\}$ be a collection of closed subsets of X. Then $\bigcap_{i \in I} U_i$ is a closed subset of X.
- Let $U_1, \dots, U_n$ be closed subsets of X. Then $\bigcup_{i=1}^n U_i$ is a closed subset of X.

Proof
Suppose that $\{U_i \mid i \in I\}$ is a collection of open subsets of X. Let $x \in \bigcup_{i \in I} U_i$. Then there exists $i \in I$ such that $x \in U_i$. Since U_i is open, there exists $r > 0$ such that $B_r(x) \subseteq U_i$. Then $B_r(x) \subseteq \bigcup_{i \in I} U_i$ and so we are done.

Suppose that $U_1, \dots, U_n$ are open subsets of X. Let $x \in \bigcap_{i=1}^n U_i$. Then for $1 \leq i \leq n$, there exists $r_i > 0$ such that $B_{r_i}(x) \subseteq U_i$. Let $r = \min\{r_1, \dots, r_n\}$. Then for $1 \leq i \leq n$, we have $B_r(x) \subseteq B_{r_i}(x) \subseteq U_i$. Hence $B_r(x) \subseteq \bigcap_{i=1}^n U_i$. Thus $\bigcap_{i=1}^n U_i$ is an open subset.

Suppose that $\{U_i \mid i \in I\}$ is a collection of closed subsets of X. Then $\{X \setminus U_i \mid i \in I\}$ is a collection of open subsets of X. Hence

$$\bigcup_{i \in I} (X \setminus U_i) = X \setminus \bigcap_{i \in I} U_i$$

is an open subset of X. Hence $\bigcap_{i \in I} U_i$ is a closed subsets of X.

Suppose that $U_1, \dots, U_n$ are closed subsets of X. Then $X \setminus U_i$ for $1 \leq i \leq n$ are open subsets.

Hence
$$\bigcap_{i=1}^{n}(X \setminus U_i) = X \setminus \bigcup_{i=1}^{n} U_i$$
is an open subset of X by the above. Hence $\bigcup_{i=1}^{n} U_i$ is a closed subset of X. ■

2.2 The Interior, Boundary and Closure Operators

Definition 2.2.1 (Interior and Boundary Points)
Let X be a metric space. Let $U \subseteq X$ be a subset. Let $x \in X$.
- We say that x is an interior points of U if there exists $r > 0$ such that $B_r(x) \subset U$.
- We say that x is a boundary point of U if for all $r > 0$, we have
$$B_r(x) \cap U \neq \emptyset \neq B_r(x) \cap (X \setminus U)$$

Definition 2.2.2 (The Interior, Boundary and Closure Operator)
Let X be a metric space. Let $U \subseteq X$ be a subset.
- Define the interior of U to be
$$U^\circ = \{x \in U \mid x \text{ is an interior point of } U\}$$
- Define the boundary of U to be
$$\partial U = \{x \in U \mid x \text{ is a boundary point of } U\}$$
- Define the closure of U to be
$$\overline{U} = U \cup \partial U$$

Proposition 2.2.3
Let X be a metric space. Let $U \subseteq X$. Then the following are true.
- U is open if and only if $U = U^\circ$ if and only if $U \cap \partial U = \emptyset$.
- U is closed if and only if $U = \overline{U}$.

Proof ■

2.3 Limits of Functions

Definition 2.3.1 (Accumulation Points)
Let M be a metric space. Let $x \in M$. We say that x is an accumulation point of M if for all $r \in \mathbb{R}^+$, we have
$$B_r(x) \setminus \{p\} \neq \emptyset$$

Definition 2.3.2 (Limits)
Let $(X, d_X), (Y, d_Y)$ be metric spaces. Let $f : X \to Y$ be a function. Let $p \in X$ be an accumulation point. We say that $L \in Y$ is the limit of f at p if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in M$ such that $0 < d_X(x, p) < \delta$, we have
$$d_Y(f(x), L) < \varepsilon$$
In this case, we write $L = \lim_{x \to p} f(x)$.

2.4 Continuous Functions

Definition 2.4.1 (Continuity)
Let $(X, d_X), (Y, d_Y)$ be metric spaces. Let $f : X \to Y$ be a function. Let $p \in X$. We say that f is continuous at p if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x \in X$ such that $0 < d_X(x, p) < \delta$, we have $d_Y(f(x), f(p)) < \varepsilon$.

Proposition 2.4.2
Let X, Y be metric spaces. Let $f : X \to Y$ be a function. Let $p \in X$ be an accumulation point. Then the following are equivalent.

- f is continuous at p.
- For every sequence $(x_n)_{n\in\mathbb{N}} \subseteq X$ such that $(x_n)_{n\in\mathbb{N}} \to x$, we have $(f(x_n))_{n\in\mathbb{N}} \to f(p)$.
- $\lim_{x\to p} f(x) = f(p)$.
- For every open subset $V \subseteq Y$, we have $f^{-1}(U) \subseteq X$ is open.

Proof ■

Proposition 2.4.3
Let X, Y be metric spaces. Let $f : X \to Y$ be a function. Then f is continuous if and only if for every open subset $V \subseteq Y$, we have $f^{-1}(U) \subseteq X$ is open.

Proof
Suppose that f is continuous. Let $\Omega \subset V$ such that Ω is open. Then for every $p \in f^{-1}(V)$, there exists $\epsilon > 0$ such that $B_\epsilon(f(p)) \subset V$. By continuity, there exists $\delta > 0$ such that $f(B_{\delta(p)}) \subset B_\epsilon(f(p))$. This implies $B_\delta(p) \subset f^{-1}(B_\epsilon(f(p)))$. But also since $B_\epsilon(f(p)) \subset V$, we have

$$B_\delta(p) \subset f^{-1}(B_\epsilon(f(p))) \subset f^{-1}(V)$$

Since this is true for every p, $f^{-1}(V)$ must be open.

Now suppose that $\Omega \subset V$ is open imply $f^{-1}(\Omega)$ is open. Let $p \in \Omega$. Then there exists $\epsilon > 0$ such that $B_\epsilon(f(p)) \subset \Omega$. By assumption, we must have $f^{-1}(B_\epsilon(f(p)))$ is open. The fact that this is open means there exists $\delta > 0$ such that $B_\delta(p) \subset f^{-1}(B_\epsilon(f(p)))$. Then we have

$$f(B_\delta(p)) \subset B_\epsilon(f(p))$$

and we are done. ■

Lemma 2.4.4
Let (X, d_1) and (Y, d_2) be metric spaces. Let $f : X \to Y$ be a distance preserving map. Then f is continuous.

Proof Let $p \in X$. Let $\varepsilon > 0$. Choose $\delta = \varepsilon$, then for all $x \in X$ such that $0 < d_X(x, p) < \delta$, we have $d_Y(f(x), f(p)) = d_X(x, p) < \delta = \varepsilon$ so that f is continuous. ■

Proposition 2.4.5
Let X, Y, Z be metric spaces. Let $f : X \to Y$ and $g : Y \to Z$ be functions. If f and g are continuous, then $g \circ f$ is continuous.

2.5 Uniform Continuous Functions

Definition 2.5.1 (Uniformly Continuous) A map $f : X \to Y$ is uniformaly continuous if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon$$

for any $x, y \in X$.

Lemma 2.5.2 If $f : X \to Y$ is uniformly continuous then it is continuous.

2.6 Lipschitz Continuous Functions

Definition 2.6.1 (Lipschitz Continuous) Let (X, d_X) and (Y, d_Y) be metric spaces and suppose that $f : X \to Y$. We say that f is a Lipschitz map if there is a constant $K \geq 0$ such that

$$d_Y(f(x), f(y)) \leq Kd(x, y)$$

for all x, y in X.
If $Y = X$ and $K \in [0, 1)$ then f is a contraction mapping.

Lemma 2.6.2 If $f : X \to Y$ is Lipschitz continuous then it is continuous.

3 Convergence in Metric Spaces

3.1 Sequences

Definition 3.1.1 (Sequences) Let X be a metric space. A sequence in X is an ordered set of numbers $x_0, x_1, x_2, \dots$ such that they all are in X. We denote this sequence by $(x_n)_{n\in\mathbb{N}}$.

Definition 3.1.2 (Convergence) Let (X, d) be a metric space. A sequence $(x_n)_{n\in\mathbb{N}} \subset X$ a metric space is said to converge to $x \in X$ if for every $\epsilon > 0$ there exists N such that $n > N$ implies

$$d(x_n, x) < \epsilon$$

Lemma 3.1.3 (Uniqueness of Limit) Let X be a metric space. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in X. If $(x_n)_{n\in\mathbb{N}}$ converges to $x \in X$ and $y \in X$, then $x = y$.

Proposition 3.1.4 Let X be a metric space. $U \subseteq X$ is closed if and only if for every sequence $(x_n)_{n\in\mathbb{N}} \subseteq U$ that converges, it converges to some $x \in U$.

Proof Suppose that U is closed. Then $X \setminus U$ is open by definition. Let $\{x_n\} \subset U$ converge to $x \notin U$. Then $x \in X \setminus U$. By definition of convergence, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $x_n \in B_\epsilon(x)$ for $n > N$. But since $X \setminus U$ is open, there should be some ϵ such that $B_\epsilon(x) \subset X \setminus U$. In this case, we would have $x_n \in B_r(x) \subset X \setminus U$ which is a contradiction.
Suppose that the right hand side is true. Suppose for a contradiction that $X \setminus U$ is not open. Then for every $\epsilon > 0$, $B_\epsilon(x)$ is not a subset of $X \setminus U$ for some $x \in X \setminus U$. Let $y_k \in B_{1/k}(x)$ but $y_k \notin X \setminus U$. Then $y_k \in U$ and $y_k \to x \in X \setminus U$, a contradiction. ■

Proposition 3.1.5 Let X, Y be metrics spaces. Let $((x_n, y_n))_{n\in\mathbb{N}}$ be a sequence in $X \times Y$. Let $(x, y) \in X \times Y$. Then $((x_n, y_n))_{n\in\mathbb{N}}$ converges to (x, y) if and only if $(x_n)_{n\in\mathbb{N}}$ converges to x and $(y_n)_{n\in\mathbb{N}}$ converges to y.

Definition 3.1.6 (Cauchy Sequence) We say that $\{x_n\} \subset (X, d)$ is a Cauchy sequence if for every $\epsilon > 0$, there exists some N such that $d(x_n, x_m) < \epsilon$ for all $n, m > \epsilon$.

Proposition 3.1.7 Every convergent sequence is Cauchy.

Proof Let $(x_n)_{n\in\mathbb{N}}$ be a convergent sequence in a metric space X. Let $\epsilon > 0$, then from convergence we have that for $d(x_n, x) < \frac{\epsilon}{2}$ for all $n > N$ for some $N \in \mathbb{N}$. Then choosing the same N, we have that

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

thus we are done. ■

We now give the definition of a complete space in terms of Cauchy sequences.

3.2 Complete Metric Spaces

Definition 3.2.1 (Complete Spaces) A metric space (X, d) is complete if any Cauchy sequence in X converges.

Proposition 3.2.2 A subspace of a metric space is complete if and only if it is closed under a complete metric space.

Proof Suppose that X is a metric space and $U \subset X$ is a complete metric space. Let $(x_n)_{n\in\mathbb{N}} \subset U$ and that $x_n \to x \in X$. Then $(x_n)_{n\in\mathbb{N}}$ is Cauchy thus it convergence to some $y \in U$. We will show that in fact $x = y$. This is true from the fact that

$$d|_U(x_n, y) = d(x_n, y)$$

Thus $(x_n)_{n\in\mathbb{N}}$ is in fact a sequence that converges in U. This shows that U is closed.
Now suppose that U is closed under a complete metric space X. Let $(x_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in U. Then trivially it is also a Cauchy sequence in X and thus is convergent. Since U is closed, the limit is necessarily in U and thus U is complete. ■

Theorem 3.2.3 (Cantor's Intersection Theorem) Let X be a complete metric space. Let $S_1 \supseteq S_2 \supseteq \dots$ form a nested sequence of non-empty closed sets in X with the property that $\operatorname{diam}(S_n) \to 0$ as $n \to \infty$. Then

$$\bigcap_{n=1}^{\infty} S_n \neq \emptyset$$

Proof For each $N \in \mathbb{N}$, choose $x_N \in S_N$. Then for all $n > N$, $x_n \in S_N$. Thus for $n, m > N$, we have that $d(x_n, x_m) \leq \operatorname{diam}(S_n)$. It follows that $(x_n)_{n\in\mathbb{N}}$ is Cauchy. Thus $x_n \to x$ for some $x \in X$. Since each S_n is closed and $x_n \in S_N$ for all $n > N$, we must have that $x \in S_n$ for each n. Thus $x \in \bigcap_{k=1}^{\infty} S_n$ is nonempty. ■

Below are a few examples of complete spaces.

Proposition 3.2.4 $\mathbb{R}^n$ and $\mathbb{C}$ are both complete.

Proof Let $(x_k)_{k\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{R}^n$. Denote the ith component of x_k by $x_{k,i}$. Then for every $\epsilon > 0$, there exists N such that

$$\|x_k - x_m\| = \left(\sum_{i=1}^{n} |x_{k,i} - x_{m,i}|^2\right)^{\frac{1}{2}} < \epsilon$$

for $k, m > N$. In particular, we have that each individual

$$|x_{k,i} - x_{m,i}| < \epsilon$$

for $m, n > N$. Thus $(x_{k,i})_{k\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$. But we know that Cauchy sequences in $\mathbb{R}$ converges, thus $(x_{k,i})_{k\in\mathbb{N}}$ converges to $x_i \in \mathbb{R}$. Now define $x = (x_1, \dots, x_n)$, then

$$\|x_k - x\| = \left(|x_{k,i} - x_i|^2\right)^{\frac{1}{2}} < n\epsilon$$

by convergence of each individual component. Thus $(x_n)_{n\in\mathbb{N}}$ is a convergent sequence.
The proof for $\mathbb{C}$ is the same in considering $\mathbb{R}^2$. ■

3.3 The Contraction Mapping Theorem

Theorem 3.3.1 (Contraction Mapping Theorem) Let X be a nonempty complete metric space and suppose that $f : X \to X$ is a contraction. Then f has a unique fixed point, meaning there is a unique $x \in X$ such that $f(x) = x$.

Proof Let $x_0 \in X$ and define a sequence by $x_{n+1} = f(x_n)$ for $n \in \mathbb{N}$. Then we have that

$$d(x_{n+1}, x_n) \leq K d(x_n, x_{n-1}) \leq \dots \leq K^n d(x_1, x_0)$$

Then for any $k > n$, we have that

$$\begin{aligned} d(x_k, x_n) &\leq \sum_{i=n}^{k-1} d(x_{i+1}, x_i) \\ &\leq \sum_{i=n}^{k-1} K^i d(x_1, x_0) \\ &\leq \frac{K^i}{1-K} d(x_1, x_0) \end{aligned}$$

This is Cauchy since we can choose $\epsilon > 0$ such that $\frac{K^i}{1-K} < \epsilon$. Since X is complete, we have that $x_n \to x$ for some $x \in X$. Since f is continuous we have that $f(x_n) \to f(x)$. Now taking limits on

$$x_{n+1} = f(x_n)$$

we have that $x = f(x)$.
To prove uniqueness, note that if $f(x) = x$ and $f(y) = y$, then

$$d(x, y) = d(f(x), f(y)) \leq K d(x, y)$$

which implies that $(1 - K)d(x, y) = 0$. Thus $x = y$. ■

Another name for this theorem would be Banach's Fixed Point Theorem.

Theorem 3.3.2 (Picard-Lindelof Theorem) Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz continuous with

$$|f(x) - f(y)| \leq L|x - y|$$

where $x, y \in \mathbb{R}^n$. Then for any $x_0 \in \mathbb{R}^n$, the differential equation

$$\frac{dx}{dt} = f(x)$$

with initial condition $x(0) = x_0$ has a unique solution on $[-t, t]$ for any $Lt < 1$.

3.4 Cantor's Theorem

Theorem 3.4.1 If X is a complete metric space and $\{F_n | n \in \mathbb{N}\}$ is a collection of open dense subsets of X, then

$$F = \bigcap_{k=1}^{\infty} F_n t$$

is dense in X. Equivalently, if $\{G_n | n \in \mathbb{N}\}$ is a collection of nowhere dense subsets of a nonempty complete metric space X, then

$$\bigcup_{k=1}^{\infty} F_k \neq X$$

Lemma 3.4.2 The Cantor set is uncountable.

4 Paths in Metric Spaces

4.1 Basic Properties of Paths

Definition 4.1.1 (Paths in a Metric Space)
Let X be a metric space. A path in X is a continuous function $f : [a, b] \to X$.

Definition 4.1.2 (Different Types of Paths)
Let X be a metric space. Let $f : [a, b] \to X$ be a path in X.

- We say that r is simple if $f(t_1) = f(t_2)$ implies $t_1 = t_2$ for all $t_1, t_2 \in (a, b)$.
- We say that r is closed if $f(a) = f(b)$.

Definition 4.1.3 (Reparametrization)
Let X be a metric space. Let $f : [a, b] \to X$ and $g : [c, d] \to X$ be paths in X. We say that g is a reparametrization of f if there exists a continuous function $\rho : [c, d] \to [a, b]$ such that $g = f \circ \rho$.

Definition 4.1.4 (Orientation Preserving Reparametrizations) Let X be a metric space. Let $f : [a, b] \to X$ be a path in X. Let $g : [c, d] \to X$ be a reparametrization of f via the continuous function $\rho : [c, d] \to [a, b]$. We say that g is an orientation preserving reparametrization if $\rho'(t) > 0$ for all $t \in [c, d]$.

4.2 Geodesics

Definition 4.2.1 (Geodesics) Let (X, d) be a metric space. A geodesic on X is a distance preserving function $\gamma : I \subseteq \mathbb{R} \to X$.

Definition 4.2.2 (Geodesic Metric Spaces) Let (X, d) be a metric space. We say that X is geodesic if for all $a, b \in X$, there exists a geodesic $\gamma : I \to X$ from a to b.